

Examples sheet 5 – Solutions – Linear Algebra

DIAGONALISATION

1. This question is motivated by the case where $\gamma(\lambda_i) < \alpha(\lambda_i)$ for one or more indices i . In this case, A is defective and therefore not diagonalizable. However, one can still write down a similar relation to connecting A and D , although an eigenvalue factorisation of A does not exist. Using the eigenvalue equation for $i = 1, \dots, n$, we find

$$A\mathbf{v}_i = \lambda_i \mathbf{v}_i \implies [A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_k] = [\lambda_1 \mathbf{v}_1, \lambda_2 \mathbf{v}_2, \dots, \lambda_k \mathbf{v}_k].$$

We show that the term on the left is AV , while the term on the right is VD . First, recall the following definitions and notation:

$$[\mathbf{c}_j(A)]_i = a_{ij}, \quad [AB]_{ij} = \sum_{k=1}^n a_{ik} b_{kj}, \quad [A\mathbf{x}]_i = \sum_{k=1}^n a_{ik} x_k.$$

Using these definitions and notation, we find

$$[\mathbf{c}_j(AV)]_i = [AV]_{ij} = \sum_{k=1}^n a_{ik} v_{kj} = \sum_{k=1}^n a_{ik} [\mathbf{c}_j(V)]_k = [A\mathbf{c}_j(V)]_i.$$

Hence, $\mathbf{c}_j(AV) = A\mathbf{c}_j(V)$, for $j = 1, \dots, k$, i.e., the j th column of AV is $A\mathbf{c}_j(V)$. Hence,

$$AV = [A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_k].$$

Let now D be diagonal with $d_{ij} = \lambda_j \delta_{ij}$

$$[\mathbf{c}_j(VD)]_i = [VD]_{ij} = \sum_{k=1}^n v_{ik} d_{kj} = \sum_{k=1}^n v_{ik} \lambda_j \delta_{kj} = \lambda_j v_{ij} = \lambda_j [\mathbf{c}_j(V)]_i.$$

Hence, $\mathbf{c}_j(VD) = \lambda_j \mathbf{c}_j(V)$, for $j = 1, \dots, k$, i.e., the j th column of VD is $\lambda_j \mathbf{c}_j(V)$. Hence,

$$VD = [\lambda_1 \mathbf{v}_1, \lambda_2 \mathbf{v}_2, \dots, \lambda_k \mathbf{v}_k].$$

The result then follows.

2. (a) We find

$$A = \begin{bmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

- i. We have

$$A\mathbf{x} = \mathbf{0} \implies \begin{cases} 2x + 2y + 2z &= 0 \\ x + y + z &= 0 \end{cases} \xrightarrow{x=\alpha, y=\beta} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ -\alpha - \beta \end{bmatrix} \implies \ker A = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

Hence $\text{rank } A = 3 - \dim \ker A = 1$.

- ii. We first compute $p_A(t)$:

$$p_A(t) = \det \begin{bmatrix} t-2 & -2 & -2 \\ 0 & t & 0 \\ -1 & -1 & t-1 \end{bmatrix} = t[(t-2)(t-1) - 2] = t(t^2 - 3t) \implies \text{sp} A = \{0, 3\}.$$

Let $\lambda = 0$. In this case, $E_\lambda = \ker A$, so $\gamma(\lambda) = 2$ and we can use as eigenvectors the spanning set for $\ker A$ found in part i. We also note that $\gamma(\lambda) = \alpha(\lambda) = 2$.

Let $\lambda = 3$. We have

$$(3I - A)\mathbf{x} = \mathbf{0} \implies \begin{bmatrix} 1 & -2 & -2 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0} \implies \begin{cases} x - 2z = 0 \\ y = 0 \end{cases} \xrightarrow{z=1} \begin{cases} x = 2, \\ y = 0, \\ z = 1. \end{cases}$$

Hence, we have a full set of eigenvectors, since $\sum_{\lambda \in \text{sp}(A)} \gamma(\lambda) = 3$, so that the matrix is diagonalisable. The corresponding eigenvalue factorisation is

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -2 \\ 0 & 3 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

(b) Let $A = \mathbf{v}\mathbf{u}^T$.

i. We have

$$A\mathbf{v} = \mathbf{v}\mathbf{u}^T\mathbf{v} = \lambda\mathbf{v}, \quad \lambda := \mathbf{u}^T\mathbf{v}.$$

ii. We show $\ker A \subseteq U^\perp$ and $U^\perp \subseteq \ker A$.

First, let $\mathbf{x} \in \ker A$. Then

$$\mathbf{0} = A\mathbf{x} = \mathbf{v}\mathbf{u}^T\mathbf{x} = (\mathbf{u}^T\mathbf{x})\mathbf{v} \implies \mathbf{u}^T\mathbf{x} = 0 \implies \mathbf{x} \in U^\perp \implies \ker A \subseteq U^\perp.$$

Let now $\mathbf{x} \in U^\perp$, so that $\mathbf{u}^T\mathbf{x} = 0$. Then

$$A\mathbf{x} = \mathbf{v}\mathbf{u}^T\mathbf{x} = \mathbf{v}(\mathbf{u}^T\mathbf{x}) = 0 \cdot \mathbf{v} = \mathbf{0} \implies \mathbf{x} \in \ker A \implies U^\perp \subseteq \ker A.$$

We conclude $\ker A = U^\perp$. Since $\dim \ker A > 0$, $\lambda = 0$ is an eigenvalue of A , with geometric multiplicity $\gamma(\lambda) = \dim \ker A = \dim U^\perp = n - \dim U = n - 1$.

iii. We found $\text{sp} A = \{0, \mathbf{u}^T\mathbf{v}\}$, with $\sum_{\lambda \in \text{sp}(A)} \gamma(\lambda) = (n - 1) + 1 = n$, so the matrix is diagonalisable. Let $B = \{\mathbf{w}_1, \dots, \mathbf{w}_{n-1}\}$ be a basis for $U^\perp$. Then $A\mathbf{w}_i = \mathbf{0}$, so $B \cup \{\mathbf{v}\}$ is a linearly independent set of eigenvectors. Hence,

$$A = VDV^{-1} := [\mathbf{w}_1, \dots, \mathbf{w}_{n-1}, \mathbf{v}] \begin{bmatrix} 0 & & & \\ & \ddots & & \\ & & 0 & \\ & & & \mathbf{u}^T\mathbf{v} \end{bmatrix} [\mathbf{w}_1, \dots, \mathbf{w}_{n-1}, \mathbf{v}]^{-1}.$$

3. (a) Let

$$A := \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T.$$

Then

$$A\mathbf{u}_1 = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T \mathbf{u}_1 = \lambda_1 \mathbf{u}_1 (\mathbf{u}_1^T \mathbf{u}_1) + 0 = \lambda_1 \mathbf{u}_1.$$

We similarly find $A\mathbf{u}_2 = \lambda_2 \mathbf{u}_2$.

(b) We show $\ker A \subseteq U^\perp$ and $U^\perp \subseteq \ker A$.

First, let $\mathbf{x} \in \ker A$. Then $\mathbf{x} \in U^\perp$ since

$$\mathbf{0} = A\mathbf{x} = (\lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T)\mathbf{x} = \lambda_1 (\mathbf{u}_1^T \mathbf{x}) \mathbf{u}_1 + \lambda_2 (\mathbf{u}_2^T \mathbf{x}) \mathbf{u}_2 \implies \begin{cases} \lambda_1 (\mathbf{u}_1^T \mathbf{x}) = 0 \\ \lambda_2 (\mathbf{u}_2^T \mathbf{x}) = 0 \end{cases} \implies \begin{cases} \mathbf{u}_1^T \mathbf{x} = 0 \\ \mathbf{u}_2^T \mathbf{x} = 0 \end{cases} \implies \mathbf{x} \perp U.$$

Let now $\mathbf{x} \in U^\perp$, so that $\mathbf{u}_1^T \mathbf{x} = \mathbf{u}_2^T \mathbf{x} = 0$. Then

$$A\mathbf{x} = (\lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T)\mathbf{x} = \lambda_1 (\mathbf{u}_1^T \mathbf{x}) \mathbf{u}_1 + \lambda_2 (\mathbf{u}_2^T \mathbf{x}) \mathbf{u}_2 = \mathbf{0} \implies \mathbf{x} \in \ker A \implies U^\perp \subseteq \ker A.$$

We conclude $\ker A = U^\perp$. Since $\dim \ker A > 0$, $\lambda = 0$ is an eigenvalue of A , with geometric multiplicity $\gamma(\lambda) = \dim \ker A = \dim U^\perp = n - \dim U = n - 2$.

- (c) We found $\text{sp}A = \{0, \lambda_1, \lambda_2\}$, with $\sum_{\lambda \in \text{sp}(A)} \gamma(\lambda) = (n-2) + 2 = n$, so the matrix is diagonalisable. Let $B = \{\mathbf{w}_1, \dots, \mathbf{w}_{n-2}\}$ be a basis for $U^\perp$. Then $A\mathbf{w}_i = \mathbf{0}$, so $B \cup U$ is a linearly independent set of eigenvectors. Hence,

$$A = VDV^{-1} := [\mathbf{w}_1, \dots, \mathbf{w}_{n-2}, \mathbf{u}_1, \mathbf{u}_2] \begin{bmatrix} 0 & & & \\ & \ddots & & \\ & & \mathbf{u}_1 & \\ & & & \mathbf{u}_2 \end{bmatrix} [\mathbf{w}_1, \dots, \mathbf{w}_{n-2}, \mathbf{u}_1, \mathbf{u}_2]^{-1}.$$

4. (a) We have

$$p_A(t) = \det(tI_3 - A) = \det \begin{bmatrix} t-3 & -4 & -4 \\ 3 & t-3 & 1 \\ -1 & 4 & t \end{bmatrix} = t^3 - 6t^2 + 13t - 20 = (t-4)(t^2 - 2t + 5).$$

Hence, the roots of $p(t)$ are $\lambda_1 = 4, \lambda_{2,3} = 1 \pm 2i$.

The corresponding eigenvectors can be found in the usual way.

$$(\lambda_1 I - A)\mathbf{x} = \mathbf{0} \iff \begin{bmatrix} 1 & -4 & -4 \\ 3 & 1 & 1 \\ -1 & 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{0} \xrightarrow{x_3=a} \begin{cases} x_1 - 4x_2 = 4a \\ 3x_1 + x_2 = -a \end{cases} \iff \begin{cases} x_1 = 0 \\ x_2 = -a \end{cases} \implies \mathbf{x} = \begin{bmatrix} 0 \\ -a \\ a \end{bmatrix}$$

which means that the eigenspace of A associated with λ_1 is

$$E_{\lambda_1} = \text{span} \left\{ \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Let now $\lambda_2 = 1 - 2i$. We find

$$(\lambda_2 I - A)\mathbf{x} = \mathbf{0} \iff \begin{bmatrix} -2-2i & -4 & -4 \\ 3 & -2-2i & 1 \\ -1 & 4 & 1-2i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{0}.$$

Adding rows 1 and 3, we find $x_1 = -x_3$. Setting $x_2 = a$, we find

$$(-2+2i)x_3 = 4a \implies x_3 = -(1+i)a = -x_1,$$

which means that the eigenspace of A associated with λ_2 is

$$E_{\lambda_2} = \text{span} \left\{ \begin{bmatrix} 1+i \\ 1 \\ -(1+i) \end{bmatrix} \right\}.$$

Finally, the last eigenspace can be obtained by taking the complex conjugate of the expression for E_{λ_3} :

$$E_{\lambda_3} = \text{span} \left\{ \begin{bmatrix} 1-i \\ 1 \\ -(1-i) \end{bmatrix} \right\}.$$

- (b) The diagonal canonical form is the eigenvalue decomposition of A over $\mathbb{C}$:

$$V^{-1}AV = D, \quad V = \begin{bmatrix} 0 & 1+i & 1-i \\ -1 & 1 & 1 \\ 1 & -(1+i) & -(1-i) \end{bmatrix}, \quad D = \begin{bmatrix} 4 & & \\ & 1-2i & \\ & & 1+2i \end{bmatrix}$$

(c) The block-diagonal canonical form represents a decomposition over $\mathbb{R}$

$$V^{-1}AV = D, \quad V = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 1 & 0 \\ 1 & -1 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} 4 & & \\ & 1 & -2 \\ & 2 & 1 \end{bmatrix}.$$

One can check that $AV = VD$

$$AV = \begin{bmatrix} 3 & 4 & 4 \\ -3 & 3 & -1 \\ 1 & -4 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ -1 & 1 & 0 \\ 1 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 3 & -1 \\ -4 & 1 & -2 \\ 4 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 1 & 0 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} 4 & & \\ & 1 & -2 \\ & 2 & 1 \end{bmatrix} = VD.$$

ADJOINT MAPS

5. (a) We have, for all $\mathbf{v} \in V$,

$$\begin{aligned} \langle f^*(a\mathbf{w}_1 + b\mathbf{w}_2), \mathbf{v} \rangle_V &= \langle a\mathbf{w}_1 + b\mathbf{w}_2, f(\mathbf{v}) \rangle_W = a \langle \mathbf{w}_1, f(\mathbf{v}) \rangle_W + b \langle \mathbf{w}_2, f(\mathbf{v}) \rangle_W \\ &= a \langle f^*(\mathbf{w}_1), \mathbf{v} \rangle_V + b \langle f^*(\mathbf{w}_2), \mathbf{v} \rangle_V = \langle af^*(\mathbf{w}_1) + bf^*(\mathbf{w}_2), \mathbf{v} \rangle_V \implies f^*(a\mathbf{w}_1 + b\mathbf{w}_2) = af^*(\mathbf{w}_1) + bf^*(\mathbf{w}_2), \end{aligned}$$

for all $a, b \in \mathbb{F}$ and $\mathbf{w}_1, \mathbf{w}_2 \in W$. Hence f^* is linear.

(b) Assume there is another map $\hat{f}^*$ satisfying the same definition. Then, for all $\mathbf{v} \in V$,

$$\begin{cases} \langle f^*(\mathbf{w}), \mathbf{v} \rangle_V &= \langle \mathbf{w}, f(\mathbf{v}) \rangle_W \\ \langle \hat{f}^*(\mathbf{w}), \mathbf{v} \rangle_V &= \langle \mathbf{w}, f(\mathbf{v}) \rangle_W \end{cases} \implies \langle f^*(\mathbf{w}) - \hat{f}^*(\mathbf{w}), \mathbf{v} \rangle_V = 0 \implies f^*(\mathbf{w}) - \hat{f}^*(\mathbf{w}) =: \mathbf{z} \perp B_V.$$

where we choose B_V to be an orthonormal basis of V . Note that $\mathbf{z} \notin B_V$ (otherwise $\|\mathbf{z}\| = 0$, which is not possible for a basis element). Assume $\mathbf{z} \neq \mathbf{0}$. Then the set $\{\mathbf{z}\} \cup B_V$ would be linearly independent and have cardinality $n + 1$, which means B_V is not maximal. This is a contradiction, so $\mathbf{z} = \mathbf{0}$ and therefore $f^*(\mathbf{w}) = \hat{f}^*(\mathbf{w})$. Since this holds for arbitrary $\mathbf{w}$, we conclude that $f^* = \hat{f}^*$.

(c) The adjoint of f^* is a map $(f^*)^* : V \rightarrow W$ satisfying

$$\langle (f^*)^*(\mathbf{v}), \mathbf{w} \rangle_W = \langle \mathbf{v}, f^*(\mathbf{w}) \rangle_V = \langle f^*(\mathbf{w}), \mathbf{v} \rangle_V = \langle \mathbf{w}, f(\mathbf{v}) \rangle_W = \langle f(\mathbf{v}), \mathbf{w} \rangle_W.$$

Since this holds for all $\mathbf{v} \in V, \mathbf{w} \in W$, we conclude $(f^*)^* = f$.

6. (a) First, we need an orthonormal basis with respect to the inner product provided. Since $\{1, x, x^2 - 1/3\}$ is orthogonal, we need to normalise this set to obtain the required basis. We let

$$p_1(x) = 1/\|1\| = 1/\sqrt{2}, \quad p_2(x) = x/\|x\| = x/\sqrt{2/3}, \quad p_3(x) = \left(x^2 - \frac{1}{3}\right)/\sqrt{8/45}.$$

The explicit expression for f^* given in Proposition 18.5, uses the above orthonormal basis for $\mathcal{P}_2$:

$$f^*(\mathbf{w}) = \sum_{i=1}^3 \langle \mathbf{w}, f(p_i) \rangle p_i.$$

We have

$$f(p_1) = \begin{bmatrix} p_1(0) \\ p_1(1) \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \quad f(p_2) = \begin{bmatrix} p_2(0) \\ p_2(1) \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{3/2} \end{bmatrix}, \quad f(p_3) = \begin{bmatrix} p_3(0) \\ p_3(1) \end{bmatrix} = \begin{bmatrix} -\sqrt{5}/\sqrt{8} \\ \sqrt{5}/\sqrt{2} \end{bmatrix}.$$

Hence

$$\mathbf{w}^T f(p_1) p_1 = \frac{1}{\sqrt{2}}(w_1 + w_2) \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}(w_1 + w_2),$$

$$\begin{aligned}\mathbf{w}^T f(p_2)p_2 &= \frac{\sqrt{3}}{\sqrt{2}}w_2 \cdot \frac{\sqrt{3}}{\sqrt{2}}x = \frac{3}{2}w_2x, \\ \mathbf{w}^T f(p_3)p_3 &= \frac{\sqrt{5}}{\sqrt{2}}\left(-\frac{1}{2}w_1 + w_2\right) \cdot \frac{3\sqrt{5}}{2\sqrt{2}}\left(x^2 - \frac{1}{3}\right) = \frac{15}{4}\left(-\frac{1}{2}w_1 + w_2\right)\left(x^2 - \frac{1}{3}\right).\end{aligned}$$

Hence,

$$f^*(\mathbf{w}) = \frac{1}{2}(w_1 + w_2) + \frac{3}{2}w_2x + \frac{15}{4}\left(-\frac{1}{2}w_1 + w_2\right)\left(x^2 - \frac{1}{3}\right).$$

(b) We have

$$\begin{aligned}f^*\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) &= \frac{1}{2} + 0 + \frac{15}{4} \cdot \left(-\frac{1}{2}\right)\left(x^2 - \frac{1}{3}\right) = \frac{1}{\sqrt{2}}p_1 + 0 \cdot p_2 - \frac{\sqrt{5}}{\sqrt{8}}p_3, \\ f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) &= \frac{1}{2} + \frac{3}{2}x + \frac{15}{4} \cdot \left(x^2 - \frac{1}{3}\right) = \frac{1}{\sqrt{2}}p_1 + \frac{\sqrt{3}}{\sqrt{2}}p_2 + \frac{\sqrt{5}}{\sqrt{2}}p_3.\end{aligned}$$

Hence, the matrix representation of f^* with respect to the orthonormal bases considered is

$$A_{f^*} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 0 & \sqrt{3} \\ -\sqrt{5}/2 & \sqrt{5} \end{bmatrix} = A_f^T.$$

7. (a) Using the explicit expression for f^* given in Proposition 18.5, we find:

$$f^*\left(\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}\right) = \sum_{i=1}^3 \mathbf{y}^T f(\mathbf{e}_i) \mathbf{e}_i = \mathbf{y}^T \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} \mathbf{e}_1 + \mathbf{y}^T \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} \mathbf{e}_2 + \mathbf{y}^T \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \mathbf{e}_3 = \begin{bmatrix} y_2 + y_3 + y_4 \\ y_1 + y_3 + y_4 \\ y_1 + y_2 + y_4 \end{bmatrix}.$$

Note that the matrix representations of f and f^* are

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad A^T = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}.$$

(b) We need to show that $\ker A^T = (\operatorname{col} A)^\perp$. We have

$$A^T \mathbf{y} = \mathbf{0} \implies \begin{cases} y_2 + y_3 + y_4 = 0 \\ y_1 + y_3 + y_4 = 0 \\ y_1 + y_2 + y_4 = 0 \end{cases} \implies \begin{cases} y_2 - y_1 = 0 \\ y_2 - y_3 = 0 \end{cases} \implies y_1 = y_2 = y_3 = a, y_4 = -2a, \quad a \in \mathbb{R}.$$

Hence,

$$\ker A^T = \operatorname{span} \{\mathbf{u}\}, \quad \mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -2 \end{bmatrix}.$$

Now $\dim \operatorname{col} A^\perp = 4 - \dim \operatorname{col} A = 4 - 3 = 1$, since $\operatorname{col} A$ is spanned by 3 linearly independent vectors (the columns of A). Hence, $\operatorname{col} A^\perp$ is spanned by a single vector and we note that $\mathbf{u} \perp \mathbf{c}_j(A)$ for all j , since $\mathbf{u}^T \mathbf{c}_j(A) = 0$, i.e., $\mathbf{u} \in \operatorname{col} A^\perp$, so we must have $\operatorname{col} A^\perp = \operatorname{span} \{\mathbf{u}\} = \ker A^T$.

The remaining relation requires us to check that $\ker A = (\operatorname{col} A^T)^\perp$, both of which can be shown to be $\mathbf{0}$.

8. (a) We use the definition of the adjoint:

$$\langle f^*(\vec{w}), \vec{v} \rangle = \langle \vec{w}, f(\vec{v}) \rangle.$$

Let $f^*(\vec{w}) = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$. We identify u_i from the above definition:

$$u_1v_1 + u_2v_2 + u_3v_3 = w_1(c_2v_3 - c_3v_2) + w_2(c_3v_1 - c_1v_3) + w_3(c_1v_2 - c_2v_1) \implies \begin{cases} u_1 = & w_2c_3 - w_3c_2, \\ u_2 = & w_3c_1 - w_1c_3, \\ u_3 = & w_1c_2 - w_2c_1. \end{cases}$$

Hence,

$$f^*(\vec{w}) = (w_2c_3 - w_3c_2)\mathbf{i} + (w_3c_1 - w_1c_3)\mathbf{j} + (w_1c_2 - w_2c_1)\mathbf{k} = -\vec{c} \times \vec{w} = -f(\vec{w}) \implies f^* = -f.$$

(b) Using part **(a)**, $\text{im } f^* = \text{im } (-f) = \text{im } f$. By Proposition 18.7, $\text{im } f = \text{im } f^* = (\ker f)^\perp$, so that

$$\mathbb{E}^3 = (\ker f)^\perp + \ker f = \text{im } f + \ker f.$$